

swap the graphics data as and when required. Today, games are packed in two, three, four or even more CDs. You as a player would always prefer to play games without any interruption. But



New games like RTCW, demand over 700 MB of install space

what if you are constrained by storage? You would have to shift between CDs during the game, which no avid gamer likes. Since storage cost per MB has plummeted, buying a large capacity hard drive is not a costly affair.

Downloads and Demos!

Game developers release trial versions of the games before the actual release of the product. These evaluation copies are available for download on the Internet or on CDs that accompany computing magazines. Every gaming enthusiast would prefer to try out such games before actually shelling out hard cash for a retail game. Typically the trial versions of the 3D games that are released will occupy a whopping 100 to 200 MB of storage space. Trying out such games that are available for free also requires enormous storage space.

Performance

Game data processing involves swapping of data at high speeds. If data transfer rates are high and the hard disk has a high rpm, the latency would be less. This improves the sifting of data from the graphics processor to the storage device (and vice versa) at extremely fast rates

that in effect will drastically improve the performance of the game. We have hard disks of 5,400, 7,200, 10,000 and 15,000 rpm. Choose the best drive with a large capacity and high rpm that will answer both your storage requirements for games as well as performance issues.

Game consoles

Gaming consoles today are packed with hard drives for the same purpose as mentioned above—to store gaming data (levels, characters, etc). You will be able to download game evaluations on to these drives for testing. Though these are fixed drives, the future will bring in more changes where you will have removable hard drives for such consoles, which will make gameplay easier. Storage device requirements will be ever growing in the gaming segment as we keep evolving in bringing more life into next generation games.

Operating Systems

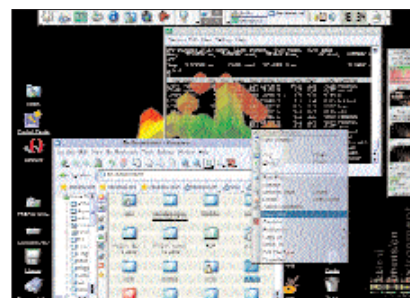
Take a look at any current generation operating system and you will find that not a single one has an install size of less than 1 GB. To top it all, Windows XP recommends a minimum install footprint of over 1.5 GB! Add general purpose programs, Office suites, along with the clutter of desktop utilities and you are struggling for free space on your 20 GB drive.

Today, personal computers have more than one operating system installed. These OSes are becoming increasingly sophisticated, as software vendors add features to their products. Likewise, the workload demand on these operating systems is growing, while users continue to demand that they be able to do more work in a shorter amount of time on their PCs. A multi-boot system would fill up a 20 GB drive in no time and for a true multi-boot system you would require multiple hard drives.

A hard drive can have as big an impact on your operating system's performance as your processor's speed. A 7,200-rpm hard drive outperforms an

Interfacing with the Future

The only way to support large and fast hard drives will be to transition from the existing Parallel ATA standard to Serial ATA (SATA). Serial ATA starts at a data transfer rate of 150 MBps and will soon migrate to 300 MBps and then to a whopping 600 MBps. It uses a thinner, more flexible cable that improves airflow and thereby facilitates the development of smaller form factor systems with better thermal designs. The evolutionary Serial ATA interface is expected to be fully adopted throughout the rest of the hard drive industry by the end of 2003. Seagate has allied closely with top innovators like LSI Logic to develop the first design-ready SATA hard drive, and to lead the SATA standards effort. The result is a hard-drive interface built for the future from the ground up, which also ensures Parallel ATA software compatibility for an easy industry transition.



With a large drive you can multi-boot between multiple Operating Systems

equivalent 5,400-rpm drive by as much as 33 per cent in system-level tests. That means the core applications of the operating system function much faster and the response time is much improved. A general way to recommend a drive capacity for your system would be to use the following formula depending on your operating

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